Predicting Healthcare Access Barriers and Identifying Population Health Profiles Using BRFSS Data

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Introduction

Access to healthcare remains a major public health concern in the United States, especially when individuals delay or avoid medical care because of cost. In 2024, 7.3% of U.S. adults reported not obtaining needed medical care due to cost, highlighting the continued presence of financial barriers to healthcare access [1]. Limited access to healthcare services is associated with reduced preventive care utilization, delayed disease detection, poorer chronic disease management and worse overall health outcomes. Although cost is a major barrier, healthcare access is also influenced by demographic, socioeconomic, behavioral, geographic, and health-related factors, making it a complex population health challenge [2, 3].

This project analyzes healthcare access barriers using data from the 2024 Behavioral Risk Factor Surveillance System (BRFSS). The primary supervised learning objective is to predict whether respondents reported needing to see a doctor during the previous twelve months but were unable to do so because of cost, represented by the BRFSS variable MEDCOST1. Identifying the characteristics associated with unmet healthcare needs may help healthcare organizations, policymakers and public health practitioners better target interventions, allocate resources and reduce health disparities.

In addition to predictive modeling, this project applies unsupervised learning techniques to identify distinct population health profiles within the BRFSS dataset. Clustering analyses provide an opportunity to explore whether respondents naturally group in subpopulations with similar healthcare access challenges, health conditions, and socioeconomic characteristics.

Together, the supervised and unsupervised learning analyses provide complementary perspectives on healthcare access. While supervised learning identifies factors associated with cost-related barriers to care and evaluates predictive performance across multiple machine learning models, unsupervised learning explores underlying population structure that may inform future public health strategies and targeted interventions.

Related Work

Several prior studies have applied machine learning methods to large public health datasets, including BRFSS, to predict health outcomes, healthcare utilization, and healthcare access barriers.

Lee et al. (2026) – Predicting Cost-Related Delays in Medical Care Using BRFSS

This paper uses BRFSS data and machine learning to predict delays in medical care due to cost while evaluating temporal generalizability and fairness across demographic groups. The authors compare several supervised learning models to identify individuals at elevated risk of delaying healthcare because of financial barriers.

Our project also predicts cost-related barriers to healthcare using BRFSS data but focuses specifically on the 2024 BRFSS dataset and complements supervised prediction with unsupervised clustering to identify distinct population health profiles.

Clark et al. (2021) – Machine Learning Using BRFSS Data to Predict Self-Rated Health

Clark and colleagues applied several machine learning algorithms to BRFSS survey data to predict self-rated health, demonstrating that machine learning can effectively model complex public health outcomes using large national survey datasets. Their work highlighted the usefulness of demographic, behavioral, and health variables for prediction.

Rather than predicting overall health status, our study focuses specifically on healthcare access barriers caused by cost. We also compare multiple supervised learning models while incorporating unsupervised clustering to explore population subgroups.

Maisog et al. (2019) – Using Massive Health Insurance Claims Data to Predict Very High-Cost Claimants: A Machine Learning Approach

Maisog et al. developed machine learning models using health insurance claims data to identify individuals likely to become very high-cost healthcare users. Their work demonstrated that ensemble learning methods can achieve strong predictive performance for healthcare resource utilization using large healthcare datasets.

Our study examines healthcare access rather than healthcare spending and uses nationally representative BRFSS survey data instead of insurance claims. In addition, we combine supervised prediction with unsupervised clustering to better characterize patterns in healthcare access barriers.

Data Source

We used the 2024 Behavioral Risk Factor Surveillance System (BRFSS) public-use dataset for this project. BRFSS is a large national health survey administered by the Centers for Disease Control and Prevention (CDC) that collects information on health behaviors, chronic conditions, preventive care utilization, healthcare access, and demographic characteristics among U.S. adults [4].

The dataset was obtained from the CDC BRFSS website and provided in SAS Transport (XPT) format. The original dataset contained 457,670 observations and 301 variables.

The supervised learning target variable was MEDCOST1, which indicates whether a respondent reported needing to see a doctor during the previous twelve months but could not because of cost.

Candidate predictor variables were selected from multiple domains:

- Demographics
- Socioeconomic status
- Healthcare access
- General health
- Chronic conditions
- Health behaviors
- Dental access
- Geographic location

After preprocessing and feature selection, the final modeling dataset contained 455,997 observations and 32 columns, including the target variable.

Feature Engineering and Preprocessing

We developed a preprocessing workflow to transform the raw BRFSS survey data into an analytical dataset suitable for supervised and unsupervised learning.

Because BRFSS contains numerous coded survey responses, preprocessing involves evaluating variable definitions, missingness patterns, skip logic, and conceptual overlap among candidate predictors.

To support consistency across project workflows, feature definitions were organized into modular feature groups within `src/feature_sets.py`. Maintaining feature definitions in a separate script ensured that all project analyses used consistent predictor definitions and allowed both supervised and unsupervised workflows to reference the same feature groups.

Candidate variables were initially selected based on domain relevance to healthcare access and cost-related barriers to care. Variables with substantial conceptual overlap were reviewed to avoid redundancy, while variables affected by survey skip logic were evaluated individually to determine whether they added meaningful information beyond related measures. Missing values were examined using BRFSS response coding conventions, including distinguishing between true missingness and survey-design-driven responses such as “Don’t Know” and “Refused.” These evaluations informed the final set of predictors retained for modeling.

The final feature set included demographic, socioeconomic, healthcare access, general health, chronic condition, health behavior, dental access, and geographic variables. Table 1 summarizes the final predictor variables retained for modeling.

Several feature-selection decisions required additional investigation. ASTHNOW was excluded because it depends on asthma-related skip logic and overlaps conceptually with ASTHMA3. Retaining both variables would have introduced redundancy while providing limited additional information. POORHLTH was retained after reviewing BRFSS coding conventions and missingness patterns because it captures activity limitations caused by poor physical or mental health and may provide insight into healthcare utilization barriers. In addition, _STATE was included to account for potential state-level variation in healthcare access environments, healthcare costs, and policy differences.

Table 1. Final Feature Categories and Variables

Feature Category	Variables
Demographic	SEXVAR, MARITAL, EDUCA, RENTHOM1, _AGE80, _RACEGR3
Socioeconomic	INCOME3, EMPLOY1, VETERAN3
Healthcare Access	PRIMINS2, PERSDOC3, CHECKUP1
General Health	GENHLTH, PHYSHLTH, MENTHLTH, POORHLTH
Chronic Conditions	CVDINFR4, CVDCRHD4, CVDSTRK3, ASTHMA3, CHCSCNC1, CHCOCNC1, CHCCOPD3, ADDEPEV3, CHCKDNY2
Health Behaviors	EXERANY2, SMOKE100, _BMI5
Dental Access	LASTDEN4, RMVTETH4
Geographic	_STATE

The preprocessing workflow was implemented in reusable notebooks and modular Python scripts to support reproducibility across both supervised and unsupervised analyses.

The final preprocessing workflow produced a modeling dataset containing MEDCOST1 as the target variable and 31 predictor variables. This dataset was exported for downstream supervised modeling and unsupervised analysis.

Part A. Supervised Learning

Methods

To predict cost-related healthcare access barriers, we evaluated multiple supervised learning model families. The three types of models tested were Logistic Regression, Random Forest, and XGBoost. The Logistic Regression model was used to represent linear models, Random Forest was used to evaluate a nonlinear model, and the XGBoost model was used due to it being known for being well suited for tabular datasets.

Logistic Regression

The Logistic Regression achieved a relatively high ROC-AUC score and a relatively high recall after threshold tuning, however it produced a lower F1 score compared to the other models. The linear nature of the model limits its ability to capture the nonlinear relationships between demographic, socioeconomic, and health variables.

Random Forest

The Random Forest model also had a good ROC-AUC score, however like the Logistic Regression model, it also had a low F1 score. In addition, it also had a low recall, meaning that it failed to identify respondents that delayed medical care due to cost. Due to this, Random Forest was not selected as the final model.

Gradient Boosting / XGBoost

Compared to the other two models, the XGBoost model uses a gradient boosting algorithm that builds decision trees to correct errors made by previous trees. Due to this, the XGBoost model had the highest ROC-AUC and F1 scores of all three models. Its overall high scores ended up being the reason why the XGBoost model was chosen.

Hyperparameter Tuning

To improve model performance on the imbalanced BRFSS dataset, several model parameters were manually adjusted based on model performance. Logistic Regression was given an increased max iteration limit and had balanced class weights. Random Forest used 400 decision trees and balanced class weights. XGBoost had 400 trees, a learning rate of 0.05, a maximum tree depth of 6, subsampling and column sampling rates of 0.8, and a positive class weight to account for class imbalance. In addition to these model parameters, the decision thresholds for each model were varied to determine which thresholds would result in the highest F1 score.

Cross-Validation Strategy

Model performance was evaluated using stratified 5 fold cross validation. Stratified sampling maintains the proportion of positive and negative responses within each fold. This produces a more reliable estimate of model performance for an imbalanced dataset. Performance metrics were averaged across the five folds and compared across all three models to determine which model to primarily use.

Overall Results

The following metrics were selected because MEDCOST1 represents a binary classification problem with potential class imbalance.

Table 2. Cross-Validated Performance Comparison of Supervised Models

	Logistic Regression	Random Forest	XGBoost	Final XGBoost
Accuracy	0.770 ∓ 0.001	0.912 ∓ 0.000	0.776 ∓ 0.001	0.873
Precision	0.255 ∓ 0.001	0.705 ∓ 0.008	0.266 ∓ 0.001	0.384
Recall	0.737 ∓ 0.005	0.133 ∓ 0.003	0.774 ∓ 0.005	0.554
F1	0.378 ∓ 0.002	0.224 ∓ 0.005	0.396 ∓ 0.002	0.454
ROC-AUC	0.836 ∓ 0.003	0.848 ∓ 0.003	0.857 ∓ 0.003	0.857

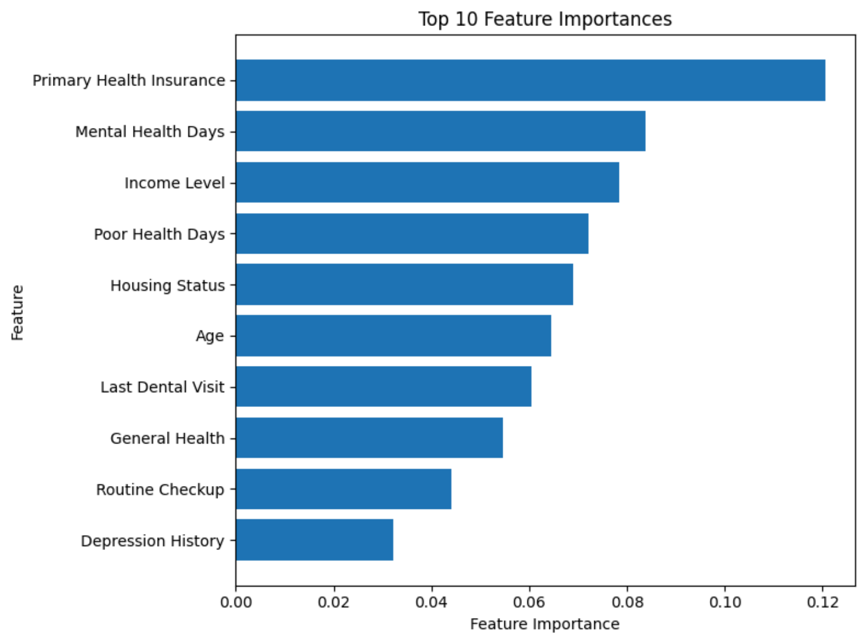
F1 score and ROC-AUC were used as the primary evaluation metrics. The F1 score provides a balanced measure of precision and recall, making it well suited for evaluating performance on imbalanced data. ROC-AUC measures each model's ability to distinguish between respondents who did and did not delay or avoid medical care due to cost across all classification thresholds. Accuracy, precision, and recall were also reported to provide additional context for model performance and to evaluate the tradeoffs between correctly identifying positive cases and minimizing false positives on the final model.

Appendix A ROC Curves for Best Models

Feature Importance and Ablation Analysis

Feature importance analysis was conducted on the best-performing model to identify predictors that contributed most strongly to model performance. From this, it was discovered that financial and general health features were considered the most important when determining if a respondent was unable to see a doctor due to cost.

Figure 4. Top Feature Importance Scores



Feature ablation analysis was performed by removing specific groups of predictors and evaluating changes in model performance. As a whole, removing groups of features hurt model performance, but certain groups caused a greater loss in performance than others. The group of features that harmed the most was the financial features, while the group that hurt the least was the chronic conditions.

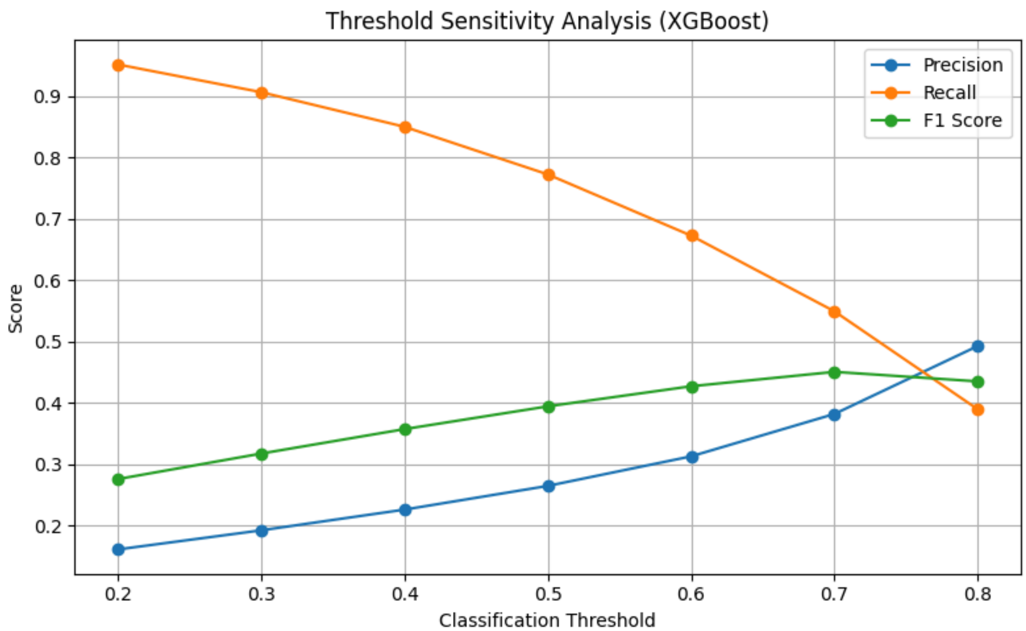
Table 3. Feature Ablation Results

XGBoost	Base Scores	Demographic Features Removed	Financial Features Removed	Chronic Conditions Features Removed
Accuracy	0.873	0.871	0.864	0.872
Precision	0.384	0.374	0.353	0.380
Recall	0.554	0.538	0.517	0.549
F1	0.454	0.442	0.419	0.449
ROC-AUC	0.857	0.847	0.834	0.856

Sensitivity Analysis

Sensitivity analysis was performed to evaluate how model performance changed under decision thresholds. Precision and F1 score began to increase as the decision threshold was raised, but the recall also began to decrease. The threshold with the highest F1 score was 0.7, so that was the final threshold used within the model.

Figure 5. Threshold Sensitivity Analysis



Tradeoffs

The supervised models had several different tradeoffs from performance to computational complexity. The most significant tradeoff was between precision and recall. Lowering the decision threshold increased recall, but decreased the precision of the model. On the other hand, raising the decision threshold increased the precision, but lowered the recall of the model. To best balance the tradeoffs between the two, the F1 score was used to best identify the threshold that best balanced the two metrics.

The other tradeoff was between computational efficiency and model complexity, which played a role in determining which model to use. Logistic regression required little training time, while Random Forest required up to fifteen minutes of training time, due to the construction of the decision trees. By comparison, the XGBoost model had a training time in between the two other models while having the highest performance. This led to the XGBoost model being the model selected for the final model.

Failure Analysis

Representative prediction failures were examined to better understand model limitations.

Failure Example 1 - False Positives

In this case, the model incorrectly predicted a cost-related healthcare barrier. The model likely assigned this high probability (0.811) due to the low income and high health risks. However, the actual response indicates there have been no cost-related barriers, for which there could be multiple reasons. Since the survey data is self reported, it is possible that there are behavioral decisions involved that the dataset cannot track.

Failure Example 2 - Missing Data

For this example, the model predicted a false negative because the respondent did not provide an entry for primary insurance type, which was the most important feature for the model in predicting healthcare barriers due to cost. It's likely that the model relied on this feature to determine if there is a barrier, and without it couldn't make an accurate decision. This same effect has been seen in other important variables as well, including income levels. The best way to improve this is to better handle missing data or make the absence of the information itself important.

Failure Example 3 - False Negatives

In the last example, the model did not predict a cost-related healthcare barrier when one existed. Despite the respondent having a low income and reporting poor health, the model still predicted there was not a healthcare barrier. The most likely reason for the misclassification is the existence of health insurance for this person, specifically Medicaid. It's likely that the model saw this person had health insurance and determined that there would be no barriers. To better address this kind of error in the future, it might help to increase sensitivity to health related features.

Appendix B Failure Analysis Summary

Part B. Unsupervised Learning

Methods

We applied unsupervised learning techniques to identify population health profiles among BRFSS respondents. The goal of using unsupervised learning was to find patterns within the respondents of the survey, specifically looking at conditions like socioeconomics, demographic information, as well as any other health-related characteristics. We want to look at these characteristics and see if we can group them together to find anything of interest.

K-means algorithm

The k-means algorithm was chosen since it's based on centroid-based algorithms that are able to break up observations into a number of groups that is represented by the number k. It then goes through each of them to find the nearest centroid that it could belong to and then keeps recalculating until the algorithm converges. It has worked well on large datasets in our research stage. Since we're working with a very large dataset here, it was something we would want to try, since it does easy-to-interpret clustering compared to other algorithms. However, k-means does have some drawbacks when doing the clustering, which is why a second method was also studied and will be used.

Gaussian Mixture Model

Gaussian Mixture Model, also known as GMM models, was the second method we chose to use, as it works in a much more different way than our first method, K-Means. GMM models focus on probabilistic approaches rather than the assignment approach that K-Means takes to the nearest centroids. GMM models work by modeling the data we use with Gaussian distributions and use the EM or expectation maximization algorithm to calculate the probability. This is very different from K-means, which does hard assignments. GMM uses soft assignment approaches. By using two different models that work very differently, we are able to understand which of these would work better on our health dataset.

Feature Representation

The features that we chose to do clustering on are from four different categories:

1. Demographics, which includes information like education, marital status, sex, home ownership, and veteran status.
2. Socio-economic background and access to care variables, such as income and number of visits.
3. Chronic health conditions. This represents things like heart attacks, stroke, asthma, and other health conditions.
4. General health behaviors and status, for example things like exercise rates, smoking history, number of poor health days, and BMI.

Altogether, these represent 30 different features per respondent in the survey. Now, before the data was fed in, a couple of things had to be done:

1. MEDCOST1 was taken out of the features entirely so that it can be used later on to validate some of the clusters that we might be testing.
2. An asthma variable was dropped since another variable already had asthma status.
3. The special codes, for example 7 or 9 for don't know/refused, 77, 88, and 99 for day count variables, were all re coded to numeric formats.
4. All the binary variables were converted into numeric, to a regular 1/0 format.
5. For the missing values, all were imputed and the median was taken.
6. They were all standardized using a scaler, which was very useful for two of the methods we used, which were k-means and GMM, since we don't want the scale of certain things like BMI to throw off all the other numbers.

After preprocessing was done for all of these, the survey data had 457,670 respondents and the 30 standardized features we would want to use when doing visualization. We used a PCA projection on all the features that we just standardized so that we can compare k-means and GMM together on the same formats. The two components that we looked at had a 22.4% difference between PC1 and PC2, which showed us that the indicators were weakly correlated

Parameter Exploration

We mainly focused on trying to select the best number of clusters or components to put in. For the first model we used, which is k-means, we tested k from 2 to 10 and did that using the elbow method as well as the silhouette scoring. We found that at k equals 2, k-means performed the best, although it only gave us a very broad split with only two clusters. For GMM, we tested this using BIC and AIC, and we found that the 10 component solution worked the best. However, this made the data too separated, and it made the data hard to understand.

Evaluation

Evaluation metrics we included:

- **Silhouette score**: This metric measures how close each survey respondent is to one another and then also to its nearest neighboring cluster. The scores in this format are represented by -1 to 1. Higher values mean that they're better spaced out and more cohesive clusters.
- **Davies-Bouldin**: This metric studies within cluster scatter versus the between-cluster separations. In this format, lower values tell us that a cluster is more compact and they're much more spaced out.

Table 4. Comparison of Unsupervised Methods

Selection Criteria	Elbow method and silhouette score	BIC / AIC
Silhouette Score	0.2056	0.0357
Davies Bouldin index	2.9572	3.758
Number of groups	2	10
Smallest / largest group size	101,434 / 356,236	7,359 / 158,697
Assignment type	Distance Based	Probabilistic
Avg. assignment confidence	Hard assignment so N/A	0.999 avg. probability, 99.7% of respondents score above 90% probability
Interpretability	2 clusters, labeled population	10 profiles find more data but are harder to understand

Cluster Visualizations

Figure 6. Cluster Visualization (Method 1)

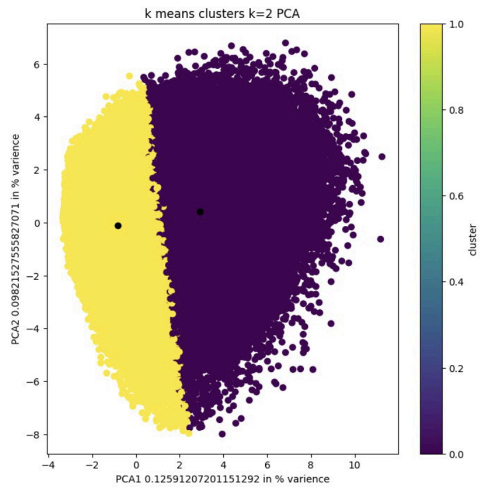


Figure 7. Cluster Profiles (Method 1)

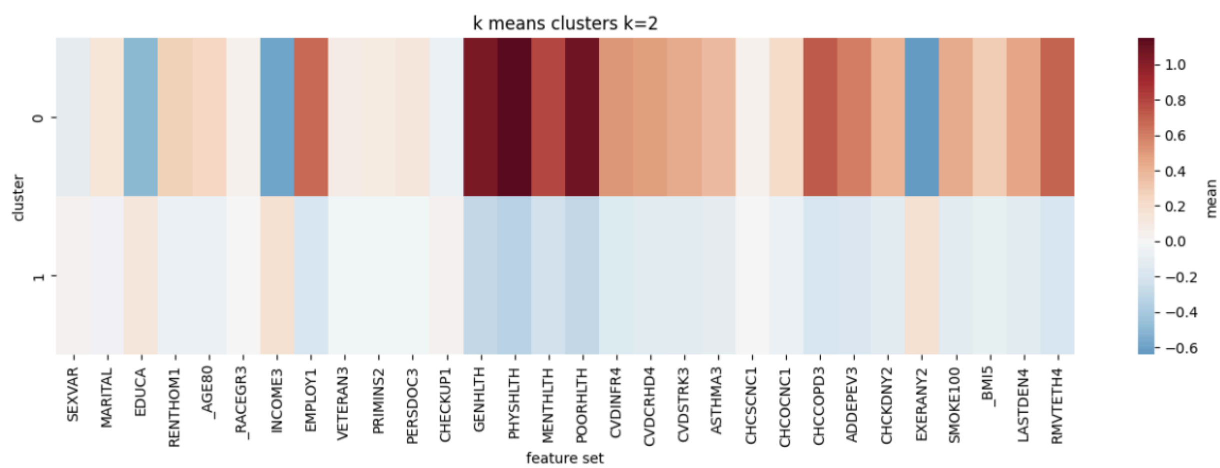
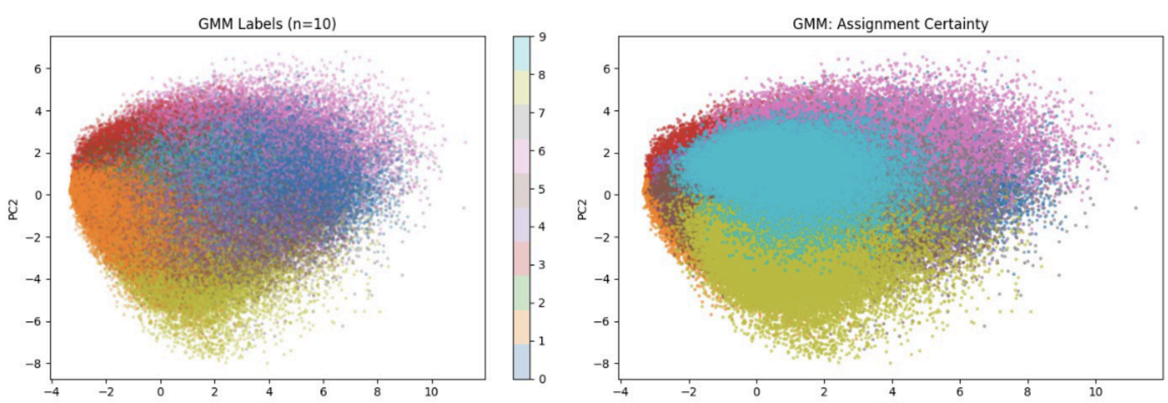


Figure 8. Cluster Visualization (Method 2)



Sensitivity Analysis

Cluster stability was evaluated across multiple parameter settings.

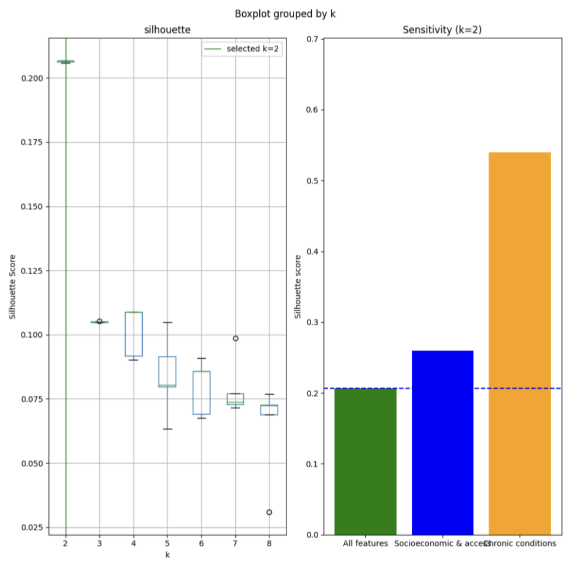


Figure 10. Cluster Sensitivity Analysis

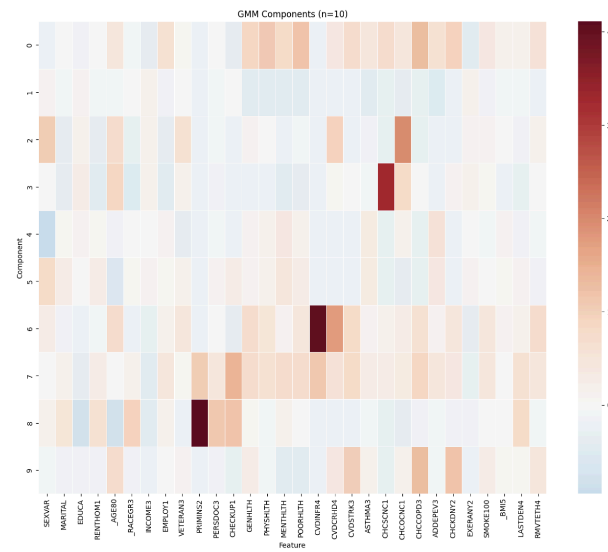


Figure 11. Cluster Profiles (Method 2)

Discussion

Supervised Learning Findings

The supervised learning models demonstrated that cost-related healthcare barriers can be predicted with relatively low success using financial and health status variables from the BRFSS survey. Among the evaluated models, XGBoost achieved the strongest overall performance producing the highest ROC-AUC and F1 score. Feature importance revealed that variables related to finances and health contributed the most to the prediction, suggesting that healthcare barriers are influenced by a combination of these factors.

Challenges

Several challenges were encountered throughout the project. One of which was the lower than average scores across all metrics, which were mainly combated through decision threshold tuning. Another challenge was the class imbalance present in the dataset which required class weighting to improve the correct identification of the target variable.

Future Work

Future work could focus on improving feature engineering, specifically on the interactions between insurance and health to improve model sensitivities. More data could also be beneficial to teach the model when health concerns can outweigh concerns about cost and vice versa.

Unsupervised Learning Findings

After doing the unsupervised learning model analysis, we found that the respondents within the survey did not separate well into well-formed subgroups, but were rather only split into two separate groups. We saw this when testing the K-Means algorithm with k equal to two. We can see the larger cluster, which had approximately 356,000 people, had characteristics such as good health, less chronic illnesses, higher income, and were mostly employed. We also see another cluster, which is smaller, that had the opposite: worse health, more chronic conditions, higher amounts of smoking, and higher amounts of unemployment.

We then wanted to validate this using medical cost, and we found that the healthier cluster had a cost barrier of 6.8%, while the lower-income cluster had 19.3%, which was double the overall average. What this shows us is that, while we can see a real substantial difference between the socio-economic and health divide, we might not be able to capture more than that using the algorithms that we tested currently.

When looking at the GMM 10 component model, we saw that there was a very big variation between the cost components, from under 2% to over 40% (not being able to afford health care). What this shows us is that, with more advanced models, we can narrow this down to find more generalizable categories. However, for now we only discovered a large difference between socio-economic groups.

Challenges

There were many challenges that we faced when doing the unsupervised learning analyses: One of the main was that there was low separation within the groups. To combat this an external label with the medical cost was used. Another issue was dealing with variables that are categorical or have specific numerical values only important to the data. For example the medical codes that were used in the data with the different types of variables that were used in this type of dataset really limited the GMM model's ability to perform to the best of its ability.

After doing some further research we realized that GMM struggles with datasets which have variables of many types, such as categorical and continuous variables, similar to how we have it in our dataset. In the future methods like k-prototypes could be used to mitigate this as they are much more suitable for datasets that contain variables of multiple types.

Future Work

In the future we aim to use other unsupervised learning models to further study which one would best perform on this type of data. Maybe study what is used in other medical surveys to really segment the population into groups rather than the broad results that we got from K-means and GMM models that we tested.

Ethical Considerations

Although BRFSS is a publicly available and deidentified dataset, ethical considerations remain important when analyzing healthcare access barriers.

For the supervised learning portion of the project, predictive models may inadvertently reinforce existing inequities if interpreted without appropriate context. Variables such as income, insurance status, geographic location, housing status, and demographic characteristics often reflect broader structural conditions rather than individual choices. Consequently, model outputs should be interpreted as indicators of healthcare access patterns rather than measures of individual responsibility.

Another important consideration is fairness across demographic and geographic groups. Model performance may differ across populations, potentially resulting in unequal error rates. Future applications of similar models should evaluate subgroup performance and avoid treating predictions as definitive indicators of healthcare need. For the unsupervised learning portion of the project, ethical concerns include overinterpreting clusters as natural or permanent categories. Clusters represent statistical patterns within the data and should be viewed as analytical tools rather than fixed population labels. Care should be taken to use neutral language and avoid stigmatizing groups that experience greater healthcare access barriers.

Statement of Work

Rechelle Transeth led the project organization, proposal development, data preparation, feature engineering, and report development. Her contributions included exploratory data analysis, candidate variable selection, preprocessing workflow development, evaluation of BRFSS coding conventions and missingness patterns, feature engineering, creation of the final modeling dataset, drafting the report framework, and integration of team contributions into the final report.

Braeden Mahnke led the supervised model development portion of the project. His contributions include developing the supervised model pipeline, selecting a supervised model, and tuning the model for performance. He also was responsible for feature importance and failure analysis on the results of the supervised model. Finally he was responsible for writing the supervised learning section of the report.

Indrayan Banerjee worked on the unsupervised learning portions, selected both the unsupervised learning models, did feature engineering to fit it to the unsupervised models, as well as doing hyperparameter tuning, as well as developing the visualizations for the unsupervised portion. Finally he worked on the unsupervised findings section of this paper.

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